TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

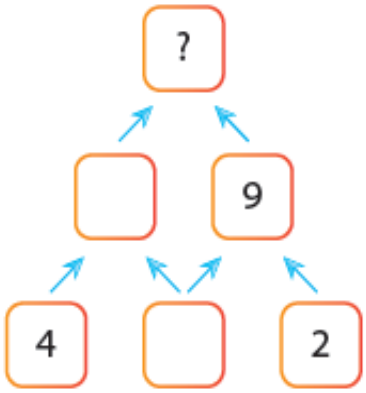
GRADE 3

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

MEDIPOL · BEYKENT UNIVERSITESI · İSTANBUL ÜNİVERSİTESİ · İSTANBUL GALATA ÜNİVERSİTESİ

1. The boxes of the number pyramid given below are filled by the rule “the sum of two adjacent numbers is written in the top box” and then some of the numbers in the boxes are deleted.



According to this what is the number in the box with question mark?

A) 18 B) 19 C) 20 D) 21 E) 22

2. At least 3 digits of a 5 digit number are less than 5 and at least 3 of which are odd digit.
What is the sum of the digits of the greatest number satisfying the above conditions?

A) 25 B) 26 C) 27 D)28 E) 29

3.

There are 50 pigeons on a wire. The sparrows are placed so that there are two pigeons on the left and right of each sparrow. Finally, owls are placed on the wire, provided that no owl is next to one another. According to this at most how many birds are there on the wire?

A) 141 B) 143 C) 145 D) 149 E) 151

4. Each of the shapes ●, ■ and ▲ represents a number. By using these shapes two equality is given below.

● + ● + ■ + ■ + ■ = ■ + ■ + ■ + ■ + ■ + ●

■ + ■ + ▲ = ▲ + ▲ + ▲ + ▲ + ■

According to given equations how many ▲ are there in the sum ● + ■ ?

A) 9 B) 8 C) 7 D) 6 E) 5

5. The sum of the digits of a four-digit number with different digits is 6. What is the sum of the greatest and the smallest value of such number?

A) 4233 B) 4320 C) 4340 D) 4350 E) 4400

6.

6. In the fgure above, a visual of the 820 m road between the home and school of Esin, who attends the 3rd grade of primary school, is given. Esin, from school and her mother, from home, starts walking towards each other along the road .

One step of Esin's mother is 60 cm and one step of Esin is 40 cm.

After having 800 steps each what is the distance in meter between them?

A) 8 B) 12 C)16 D) 20 E) 24

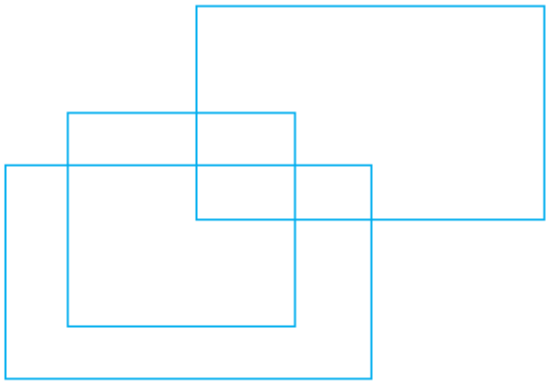
7.

Pine Hornbeam Sycamore

A district municipality has carried out afforestation activities by using 8 trees by using 3 different types of trees so that the distance between two consecutive trees of the same type is constant and the trees are on a straight line.
The distance between the first tree and the last tree is 118 m and the distance between two plane trees is 45 m, how many metres is the distance between pine and hornbeam trees?

A) 17 B) 22 C) 24 D) 27 E) 28

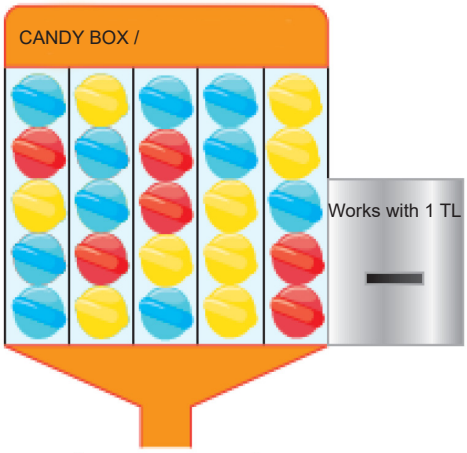
8. The figure below was obtained with the help of three rectangles intersecting at right angles.



How many rectangles are there in the figure?

A) 11 B) 12 C)10 D) 9 E) 13

9.



The figure on the left shows an automatic candy box that dispenses the bottom row of candies from a random compartment every time 1 ₺ is thrown.

Minimum, how many 1 ₺ can Bahar, who likes to eat red candies,put into the machine to guarantee at least one red candy?

A) 6 B) 10 C)11 D)12 E) 15

10. When 53 is added to or 53 is subtracted from a two-digit natural number, if the result is not two-digit, let's call that number a "Troublesome Number".

How many Troublesome Numbers are there?

A) 18 B) 17 C)16 D)15 E) 14

11. Numbers that are read the same both in front and backwards are called "Palindrome numbers".For instance 434 and 1331 are two palindrome numbers. What is the greatest value of the difference of two three-digit palindrome numbers whose sum of the digits is 13?

A) 334 B) 324 C) 243 D) 233 E) 143

12.

In a class of 34 students, there are 13 desks. The students sat in all of these desks with 2 or 3 people. According to this, in how many desks there are 2 students sitting?

A) 5 B) 6 C) 7 D) 8 E) 9

13. Among the three friends named Asena, Berçem and Cemile, Asena is 150 cm tall. The height diference between Asena and Berçem is 7 cm and the height diference between Berçem and Cemile is 5 cm.
According to this, which of the following can not be the height of Cemile in cm?

A) 138 B) 148 C) 152 D) 156 E) 162

14. Mrs Ayşe bought 4 pencils and 3 erasers from the stationery shop for her children and paid 107 ₺.

If the price of one pencil is 17 ₺ then what is the price of one eraser?

A) 9 B) 11 C) 13 D) 14 E) 15

15.

A squirrel stops 3 times while taking the walnuts it collects in the forest to its home. At each break, it eats half of the walnuts it carries and takes 12 walnuts home. According to this, how many walnuts did this squirrel eat on the road?

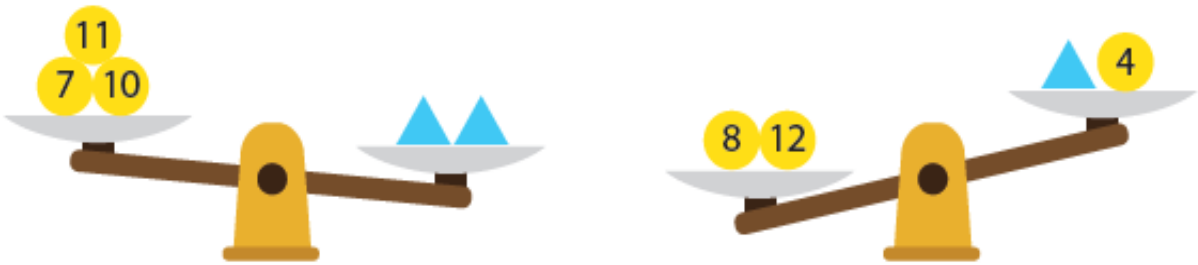
A) 96 B) 90 C) 84 D) 36 E) 24

16. The following information is given about three siblings going to 3rd, 4th and 5th grades.
Ali is not older then Aslı
Aslı not olden then Asya

According to the informations given above , which of the followings shows the order of these siblings from the youngest to the oldest?

A) Asya - Ali - Aslı
B) Asya - Aslı - Ali
C) Ali - Aslı - Asya
D) Aslı - Ali -Asya
E) Aslı - Asya - Ali

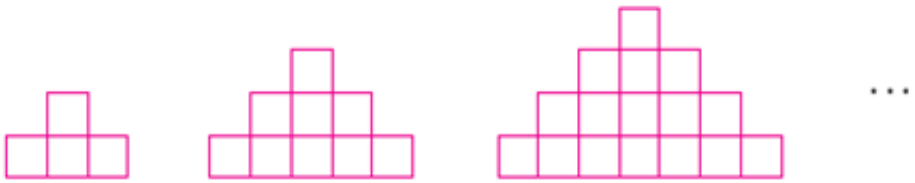
17. Two unbalanced balances are given below.



In this balances if the mass of the ▲ is a natural number, what is that number ▲ ?

A) 15 B) 16 C) 17 D) 18 E) 19

18. The first 3 steps of a pattern obtained by joining squares with a side length of 1 cm are given below.



What is the circumference of the shape obtained in the 10 th step of this pattern?

A) 60 B) 62 C) 64 D) 80 E) 92

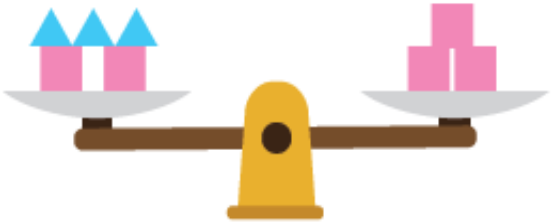
19. A balance in equilibrium is given below.

According to this, which of the following balance cannot be in equilibrium?

A)

B)

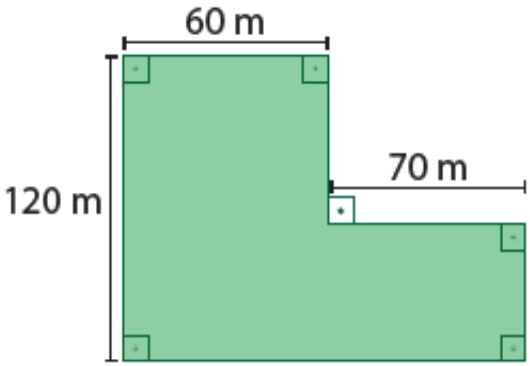
C)



D)

E)

20. A sketch of a park is given below.



How many metres does Ali run if he rans 3 laps around this park?

A) 750 B) 1000 C) 1250 D) 1500 E) 1750